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Automatic quality compliance checking in concrete dam construction: Integrating rule syntax parsing and semantic distance

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ABSTRACT

The compliance checking of concrete dam construction quality often relies on the manual reference of specifications, which is time-consuming, labor-intensive, and prone to errors. Automating compliance checking is an effective means of ensuring the quality of dams. However, using computers to match and compare quality record texts with specification provisions remains challenging. Due to the numerous specifications and diverse forms of constraints in dam construction, as well as unstructured and non-standard quality records, semantic differences exist between quality records and specifications that prevent timely quality control, making it necessary to develop a complete and efficient construction quality compliance checking framework. To address these issues, an automatic construction quality compliance checking method based on rule syntax parsing and phrase semantic distance was proposed in this work. First, key entity knowledge was determined and automatically extracted. Then, syntax parsing rules were defined, and the parsing algorithm was used to assemble the key entities into syntax trees. Then, specification and quality record syntax trees were aligned based on node semantic distance. Finally, a scoring process was designed to achieve automatic compliance checking of concrete dam construction quality. The experiments showed that the entity knowledge extraction F1 value of the proposed method was 12.89% higher than similar models, and the checking accuracy was 88.89%, with excellent applicability to both quantitative and qualitative specification constraints. The proposed method constructed a complete quality compliance checking framework for the construction field, automatically obtaining precise scores as checking results, saving considerable time compared to manual checking, and promoting compliance checking throughout the entire lifecycle of construction.

1. Introduction

Concrete dams account for a large proportion of 100-meter-high dams worldwide, significantly contributing to the national economic development of numerous countries by providing flood control, power generation, irrigation, navigation, and water supply services. In the event of dam failure, the rapid release of reservoir water in a short period can cause enormous losses to the national economy, as well as pose a threat to human life and property. Therefore, ensuring the construction quality of concrete dams during the construction period is critical for preventing disasters. Construction quality compliance checking serves as a crucial step in construction quality management.

This involves checking whether construction operations comply with the specification constraints and upholding project quality.

In hydraulic engineering, the typical practice for assessing construction quality often involves manual operation procedures. The quality inspectors compare the sampling inspection results of the construction site with the specification text and their own engineering knowledge to determine if the construction quality complies with the standards [1,2]. These on-site construction quality sampling results are documented and stored in the form of unstructured text. Due to the extensive scale of concrete dam engineering, quality records often originate from multiple different management personnel, resulting in differences in how the text is expressed. Moreover, records may contain

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both numerical quantitative and linguistic qualitative descriptions, with complex information, thus increasing the difficulty of quality compliance checking. In terms of specifications, numerous types of concrete dam construction specifications serve as checking principles, with each specification containing numerous clauses, and some clauses containing complex semantic structures, which are difficult to directly convert into usable information. During compliance checking, management personnel need to analyze the semantics of quality records, identify key information, and carefully review the specification clauses to determine whether the on-site testing results are compliant. This process is inevitably influenced by the subjective judgment of managers, which can hinder the timeliness and accuracy of quality checking. In addition, frequent specification revisions can further increase the difficulty of manual checking. Compliance checking relying solely on manual methods is time-consuming and error-prone, which brings risks to the production and lives of residents and makes it difficult to meet the high efficiency and profit requirements of construction units [3]. Therefore, automating the implementation of concrete dam construction quality compliance checking offers significant practical value.

Over the last few decades, researchers have explored methods to improve the efficiency of compliance checking, with early research relying on the use of hard-coded methods to implement compliance checking using manually written rules, which have been applied to steel structures [4], building envelopes [5], and sustainable construction [6]. This has led to the development of corresponding platforms for compliance checks [7]. However, this hard-coded checking method based on manual construction rules is relatively mechanized, and coding results are challenging to maintain. To automatically the extraction of design information, researchers have introduced the ontology concept [8] and Building Information Modeling (BIM) technology [9,10] for compliance checking, where the knowledge representation capabilities of ontology can be combined with the design information of BIM to identify architectural designs that deviate from established norms [11,12]. Ontology can be used to connect specification rules and Industry Foundation Classes (IFC) files that contain design information to perform compliance checks through SPARQL functions or logical reasoning [13–15]. However, construction is a dynamic process, and daily construction progress is challenging to quickly incorporate into the BIM model. In fact, the quality testing data of dam construction is presented in the form of virgin text records. To automatically extract key information from text, Natural Language Processing (NLP) technology has been applied to automated compliance checking as a link between computers and human languages. NLP has played a significant role in automatically extracting normative constraint information [16,17], extracting constraint relations [18], and converting normative constraints into computer-executable codes [19,20], marking a significant step forward in the research of automated compliance checking.

The above efforts have improved the flexibility of expressing regulatory knowledge and reusing constraint information, providing a solid foundation for investigating the quality compliance checking of concrete dam construction. However, limitations still persist that cannot fully address quality compliance evaluation during the construction phase. Specifically, this is particularly evident in the following aspects. (1) Existing research has focused more on BIM design checking [13,14,21–24], because BIM models could be assisted in design review with tools such as Revit Model Checker [25], but construction quality checking lacks such auxiliary tools. In addition, current approaches to compliance evaluation often yield a binary outcome of either compliance or non-compliance [15,20,26], and there were few checking methods for qualitative constraints. However, the construction quality information relies on human writing, due to diverse recording habits, original text records of construction quality often blend both quantitative and qualitative descriptions. For a construction quality record, situations may arise where half of the sentences comply with regulatory constraints, while the other half violates regulatory constraints. In such cases, evaluating compliance checking cannot be restricted to

straightforward compliance/non-compliance definitions, and instead necessitates more refined measurements that reflect compliance, such as a scoring system. (2) BIM models represent objects using classes and attributes [26], and design information is stored in a semi-structured form. With the help of class indexing, reasonably complete object information can be obtained. However, quality information on-site was implicit in text records and had no clear boundaries with surrounding text. Furthermore, specifications often include multiple branch constraints and have complex semantics [24], with an average machine interpretability of only 34.4% [27]. Implementing quality compliance checking requires extracting information from both specification texts and on-site records simultaneously, demanding higher precision in extraction and the development of more competitive information extraction models. (3) Due to the large number of recording personnel, vague and informal expressions with similar semantics but different expression methods may be used [28]. The commonly used unsupervised semantic alignment methods [14,22] struggle with handling out-of-vocabulary words in oral records.

To address these issues, an automated method for checking the quality compliance of concrete dam construction based on existing research was proposed. The main contributions of this method were as follows: (1) We designed a compliance checking framework for the construction phase, which replaced binary conclusions (compliance/non-compliance) with a scoring system to achieve a detailed compliance evaluation of the qualitative and quantitative construction quality. (2) A pre-trained model was combined with contextual semantics to improve information extraction accuracy, fusion model can deeply learn quality records and specification text features, as well as automatically extract quality records and specification constraints with rich language meanings. Syntax parsing algorithms were used to parse quality records and constraint information into check information trees and specification trees, respectively. Compliance check tasks were then implemented by branches, which reduced human intervention for normative interpretation. (3) A semantic distance model based on deep learning was selected to solve the problem of out-of-vocabulary words, this model is more competitive in semantic differentiation and could be used to align branch nodes of the check information tree and the specification tree to identify the most suitable checking rule, and the semantic distance was used instead of regular matching to implement qualitative constraint compliance checks. The proposed method expanded the application of automatic compliance checking during the construction stage. By inputting the original text records, the compliance score of each record could be automatically obtained, which was conducive to ensuring project quality and improving the efficiency of construction quality management.

The remaining sections of this manuscript are organized as follows: Section 2 reviews existing research achievements and analyzes research gaps. Section 3 introduces the principle and framework of the proposed method. Section 4 elaborates on and analyzes the application results. Section 5 discusses the main advantages and significance of the proposed method. Finally, Section 6 summarizes the contributions and limitations of this study.

2. Related studies

With societal development, communities and governments have increasingly put forward higher requirements for the quality of engineering design and construction. As a result, automated compliance checking has attracted attention from researchers. To achieve automated compliance checks for construction quality, the first step is to extract constraint information and on-site construction information from the text and interpret them as rules that computers can understand. The next step is to align the information to identify the correct constraint rules for reasoning. Finally, a comprehensive system that encompasses the aforementioned steps needs to be designed for checking. Hence, we reviewed current research related to automated compliance checking in

three categories: (1) Checking information extraction and interpretation; (2) Content alignment of rules and records; and (3) Whole-process compliance checking. The first two categories serve as critical components for realizing automated compliance checking, while the last category signifies the complete realization of the entire compliance checking process.

2.1. Checking information extraction and interpretation

Researchers in the engineering field have increasingly explored information extraction and representation methods in energy-saving codes, quality standards, and building specifications [20,29–31]. Hjelseth and Nisbet [32] semi-automatically implemented the extraction of regulatory constraints by annotating keywords in regulatory statements using four types of rules: requirement, applies, select, and exception (RASE). With the development of NLP technology, automatic information recognition and extraction methods [33,34] based on deep learning have been used in the construction industry. El-Gohary et al. [17,35,36] focused on the extraction of construction specification content and specification interpretation, while Zhong et al. [37] developed a prototype system to automatically extract constraint information from construction quality-checking procedures. Guo et al. [38] evaluated the text classification method for the intelligent review of construction organization design of water conservancy projects to control the integrity and compliance of construction. Tian et al. [39] excavated concrete dam construction knowledge using NLP technology and mutual information theory and established an information database. In addition, El-Gohary et al. used ontology semantics and deep learning algorithms to extract key construction information in checking reports [40] and the construction safety relationship of safety specifications [41], in addition to exploring the use of semantic logic to represent regulatory constraint rules and thereby creating conditions for automated reasoning.

In terms of rule interpretation, Salama et al. [42] introduced a deontic theory in engineering, which combined deontic logic with NLP to express the obligations, prohibitions, and licenses in specifications. This approach also improved the interpretation capacity of specifications. Dimyadi et al. [43] proposed a criterion for specification interpretation, which could encode regulatory knowledge into a proposed computable representation, and provide a set of regulatory knowledge coding guidelines. Zhong et al. [44] transformed specification constraints into Ontology Web Language axioms and Semantic Web Rule Language rules, allowing for the expression of specifications as regular checking tasks. Kim et al. [23] used visual language to convert specifications into computer-executable code, offering a new approach for implementing compliance checking for users unfamiliar with computer programming. Zhang and El-Gohary [24] used natural language generation techniques to directly represent canonical semantic information as intelligent code that could be executed by computers, laying the foundation for downstream compliance inference tasks.

Research on the extraction and interpretation of specification rules has significantly supported the automatic extraction of normative information. However, the specification provisions used as judgment standards serve as only one of the two parties involved in compliance checking. Research on rule interpretation marks the initial stage of overall compliance checking, intending to aid computers in understanding the specification content. However, it has failed to achieve complete compliance checking on its own.

2.2. Content alignment of rules and records

Compliance checking comprises two main components: the information to be assessed and the specification rules. To achieve effective content alignment, the concepts and relationships from both components must be transformed into the same expressions, allowing for rule-based reasoning to identify cases that deviate from the rules. Early alignment and reasoning require the manual development of rules;

therefore, Pauwels et al. [45] manually developed mapping rules to map specification terms to BIM terms, achieving content alignment. Moreover, Dimyadi et al. [46] manually translated BIM information and specifications into a specified regulatory knowledge query language to achieve content alignment and reasoning, while Melzner et al. [47] developed coding rules to achieve content alignment by manually translating specifications into BIM language, to aid in the early identification of fall hazard sources in safety design. However, this manual approach is time-consuming and difficult to maintain. Semantic vectors provide direction for automated content alignment, as similar linguistic descriptions have similar semantic vectors [48]. Xu et al. [49] successfully performed the sentiment analysis of internet public opinion by representing emotional text with semantic vectors, while Tian et al. [50] used semantic vectors to represent concrete dam construction text terms to achieve the rapid identification of large-scale construction hazards. Furthermore, Zheng et al. [14] used unsupervised learning methods to obtain vector representations of entity words in a public dataset for content alignment and used the GraphDB inference engine to perform compliance checks on BIM design files. Zhou and El-Gohary [51] proposed a domain ontology alignment method, which combined the concept of matching and unsupervised learning based on the knowledge of the construction field, and could completely automatically align BIM information with energy code terms to conduct rule reasoning.

The above studies can serve as a valuable reference for realizing content alignment and rule reasoning for concrete dam construction quality. However, rule-based content alignment and rule inference methods require considerable manual intervention. Unsupervised alignment methods do not require manual labeling of the dataset, but their semantic representation capability is significantly related to the training dataset size, making it challenging to address the issue of out-of-vocabulary terms in specialized terminology. Furthermore, while existing methods excel in extracting normative quantitative constraints and qualitative constraints, there remains a research gap in carrying out compliance reasoning for qualitative constraints. Qualitative constraints lack a clear numerical relationship, making it necessary to start the analysis from a semantic level to determine whether the real records on site are consistent with the normative constraints.

2.3. Whole-process compliance checking

The objective of many research studies involves the complete realization of automatic compliance checking. Most existing compliance checking systems, such as the Solibri system, FORNAX system, and the SMARTcodes platform, require a large number of manual rules or establishing a mapping table [7], increasing the difficulty of platform maintenance. To improve the automation level of compliance checking, Boukamp et al. [52] and Nguyen et al. [53] proposed a framework for the automated processing of regulatory content, by analyzing in detail the functions that compliance evaluation systems should contain, along with possible implementation methods. Macit and Günaydın [22] manually extracted formalizable normative concepts and entity terms individually, and used the semantic-oriented paradigm to improve the RASE method. This enhancement allowed for compliance/non-compliance warnings to building design. Choi et al. [26] designed an automated compliance checking system, which interpreted regulations as computer-understandable rules through hard coding, and combined the BIM model to conduct the compliance checking of building evacuation design. The above research reduced the degree of manual participation in the compliance checking process, however, it did not completely eliminate human involvement.

Many studies have attempted to achieve comprehensive automated checking. Zhang et al. [15] integrated NLP techniques and logic programming methods into a system, utilizing the B-Prolog platform to automatically check compliance by comparing the logical rules extracted from specifications with logical facts extracted from BIM. Xu et al. [54] expanded the ability of the SPARQL function to represent BIM

spatial rules, proposed a spatial logic reasoning mechanism for underground public facilities, and obtained compliance/non-compliance checking reports combined with third-party reasoning API. Guo et al. [13] proposed a method to automatically generate SPARQL queries, using custom term dictionary matching and semantic similarity analysis methods to implement content alignment and provide automatic rule interpretation. Xue and Zhang [55] developed a set of part-of-speech tagging algorithms tailored to building codes, which could automatically extract code requirements and expand the scope of codes used for automated checking. Zheng et al. [14] proposed a novel knowledge-based compliance checking framework, utilizing ontology and unsupervised semantic alignment to enhance the interpretation process of complex normative rules, accelerating the speed of compliance checking based on IFC files. Zhou et al. [25] proposed an innovative rule interpretation framework based on context-free grammar. This method used a pre-trained model to label the specification phrase, utilizing a syntax tree to represent the specification hierarchy, and a revit checker to provide compliance/non-compliance results for BIM fire protection design. This method was mainly aimed at BIM design checking with IFC data as the checking object, resulting in limitations for compliance checking in the construction stage. As previously explained, the dynamic nature of the construction process makes it difficult for the BIM model to participate. Manually recorded quality results may include both quantitative and qualitative descriptions. The degree of compliance between qualitative descriptions and specifications includes multiple levels, rendering the binary compliance/non-compliance inadequate for assessing its compliance effectively. Also, both construction quality results and specification provisions come from plain text, necessitating an information extraction model with enhanced generalization capabilities to extract quality and constraint information from these divergent format texts. In addition, how to select the correct branch of the specification provisions from the semantic level, and how to measure the difference between the qualitative quality description and the specification constraints from the semantic level need further study.

According to the above literature, the compliance checking methods have mainly focused on the design field, such as fire protection design, and energy-saving design. The scope of use of these methods was limited by the IFC files. However, in terms of achieving compliance checking of the equally important construction operation stage, progress remains in the exploration stage. Unlike the design stage, specific construction information during the construction stage is difficult to quickly and structurally reflect in information models such as BIM, resulting in a lack of support from IFC files, making it difficult for existing methods to directly migrate and use. In addition, situations may occur in which the preceding paragraph text of the record conforms to the specifications, while the latter paragraph violates the specifications. Therefore, the compliance degree of quality records is difficult to define solely through a binary judgment approach of compliance/non-compliance, and a more refined compliance checking method is needed for the construction stage.

2.4. Knowledge gap in automatic construction quality compliance checking in concrete dams

Automatic compliance checking involves multiple engineering sub-fields, with a unified consensus among researchers regarding the overall steps in implementing compliance checking. However, most studies have focused on extracting normative content and rule interpretation for checking preparation. Only a few studies have assessed fully implementing construction quality compliance checking, with the remaining studies focusing on checking design files. Design files are supported by BIM-based design information, with a unified IFC standard facilitating rule checking. However, quality information during concrete dam construction is mainly recorded in textual format, with the presence of ambiguity and colloquial expressions, making it difficult to align with regulatory texts and challenging compliance checking. Consequently,

there remains a lack of widely accepted methods for the compliance checking of concrete dam construction quality.

From an automation perspective, as deep learning technology advances, research is shifting from manual normative information extraction, involving human participation, to semi-automated extraction [21,56] and fully automated extraction [15,57], based on ontology and NLP. However, information extraction only constitutes one step in compliance checking, and more research is needed to attain fully automatic compliance checking of concrete dam construction quality.

In terms of research objects and methods, a gap exists in terms of the compliance checking framework that can be directly used in the construction stage, utilizing original text records as the objects under examination. Construction sites have numerous quality checking personnel, leading to diverse expressions of quality records. Moreover, construction specifications often contain many provisions and complex semantics, making it necessary to find a more powerful text knowledge extraction model that can capture key information from records and specifications. In addition, most studies have focused on quantitative checking, for qualitative constraints, it is difficult to define compliance degree solely based on the binary judgment approach of compliance/non-compliance. Therefore, it is necessary to explore more detailed compliance representation methods.

Based on the above situation, it can be seen that the existing methods struggle to meet the needs of construction quality compliance checking due to: (1) Extracting information from on-site records and normative texts requires a high-precision model. (2) Aligning colloquial records with normative provisions and performing compliance checking on qualitative quality records requires a high discriminative capability semantic alignment algorithm. (3) The method of describing the degree of compliance with binary compliance/non-compliance is more suitable for structured design checking rather than construction quality checking. Consequently, this study aims to propose an automated compliance checking framework applicable to the construction phase. This research amalgamates the advantages of multiple information extraction models to enhance extraction precision, employs deep learning-based semantic similarity algorithms to align colloquial expressions with normative constraints, and concurrently uses this algorithm to score qualitative quality descriptions. Finally, a compliance scoring system is designed to evaluate the compliance of quality texts containing qualitative, quantitative, and mixed records from the perspective of scoring, to improve the efficiency of construction quality management.

3. Proposed methodology and framework

To realize automated compliance checking during the construction phase, this study proposed a checking framework consisting of four steps, as follows: (1) Candidate specification retrieval was conducted, which retrieved the most valuable candidate specification based on the quality record through semantics; (2) Checking information extraction, which used a deep learning model to automatically identify and extract the key checking information of the quality records and the candidate specifications; (3) Checking information parsing, which represented parsing the information of quality records and candidate specifications into computer-processable forms through defined representation rules; and (4) Compliance checking implementation, which first aligned the field records for use with the specification terminology, and then designed scoring rules to score the compliance.

Many relevant specifications exist for concrete dam construction, each containing multiple constraint clauses. Throughout the construction process, providing suitable responses to the construction operations noted in the specifications is necessary. During self-inspections or supervision inspections carried out by the construction unit, quality specifications must be meticulously followed according to the provisions in the specifications, and details inspection records must be accurately maintained. Based on this correlation, several potential reference specifications clauses can be extracted from the specification library,

according to the semantics of quality records. This operation can greatly reduce the number of specifications required for checking and facilitate a streamlined response to updates by simply updating the corresponding specification clauses in the specification library. This strategy is advantageous for future maintenance efforts. Specific retrieval methods can be found in the research conducted by Zhang et al. [58]. This study, however, focused on the compliance checking operations after obtaining candidate specifications.

3.1. Checking information extraction network

Concrete dam construction specifications have a unique language expression structure. By analyzing the significant knowledge information within the norm clauses, we found that the complete specification semantic expression followed a similar sentence pattern. This pattern typically included zero or more preconditions (construction conditions, applicable conditions of the provisions), constraint objects (guide wall, fly ash, and curtain grouting), zero or more object attributes (slump, drill location, length, and width) or operations (installation, inspection method), constraint words (should, should not, would, and would not) or comparative words (greater than, less than, not more than, and not less than), and specific constraints (attribute values, operation methods). The expression form of the specification could be abstracted into a general form, as shown in Fig. 1.

Starting from the specification usage conditions, the top row in Fig. 1 represents the constraint main line. The condition could be a specific situation, such as “the use of alkali-aggregate” or a sentence with constraints and constraint values, such as “hot-rolled steel bar with a diameter greater than 10 mm”. Under this condition, the constrained objects were subjected to constraint restrictions. The constraint object possibly contained one or more object attributes, and each constrained attribute possibly contained constraint restrictions and values.

For the convenience of representation and computer recognition, in this study, we referred to the set of semantic information elements proposed by Zhang and El-Ghory [59] to represent key entities, which contained eight elements, namely, the subject, compliance checking attribute, deontic operator indicator, comparative relation, quantity value, quantity unit, subject relation, and reference. In the context of construction specification, specific choices based on this set were made. As shown in Table 1, “obj”, “aobj”, “req” and “val” corresponded to the subject, compliance checking attribute, comparative relation, and quantity value in the set of semantic information elements, respectively. In addition, three types of information were supplemented, with the object condition (con) referring to the specification usage condition, the attribute condition of the object (acobj) in the label serving as a supplementary explanation of the object attribute, and the attribute of requirement value (aval) was a supplementary explanation of the constraint value. For example, “for hot-rolled steel bars with a diameter

Table 1
Mapping of key entity knowledge categories [59].

Name	Class label	Example
Constraint object	obj	Concrete, form, watering time
Object condition	con	Prefabricated components, alkali-aggregate
Attribute of object	aobj	Wall panel height, air content, strength
Attribute condition of the object	acobj	Severe cold area, standard deviation of compressive strength
Request	req	Should be greater than, would be, less than, not exceed than
Requirement value	val	4.0 MPa, 2.0 m, 75%, 2 times
Attribute of requirement value	aval	Design strength, welded bar diameter

greater than 10 mm, the weld height of steel bars and steel plates welding should not be less than 0.35 times the diameter of the steel bars to be welded”, the constraint object is “hot-rolled steel bars”, the object condition is “diameter greater than 10 mm”, the object attribute is “weld height”, the constraint word is “should not be less than”, and the constraint value is “0.35 times”. In addition, the object attribute “weld height” is supplemented by the object attribute condition “steel bars and steel plates welding”, and the constraint value “0.35 times” is supplemented by the constraint value attribute “diameter of the steel bars to be welded”, to obtain the complete meaning of the specification.

Entity knowledge in the text can be used as checking information for implementing compliance checks. The extraction of checking information can be divided into two stages: the first is to identify the boundary between checking information and irrelevant information and obtain complete checking information, and the second is to predict the category of the checking information. To obtain complete checking information, it is necessary to understand the semantic context of the text correctly, and break before and after the complete entity knowledge to ensure the integrity of the checking information. The expressions in construction quality records are diverse, and the semantics of specification provisions are complex, increasing the difficulty of text feature extraction. This made it necessary to use a powerful deep learning model to extract checking information to improve accuracy. Bidirectional Encoder Representations from Transformers (BERT), a deep pre-training model developed by Google AI Research Institute, has been trained on massive amounts of data, utilizing three types of vectors—word embedding, sentence block, and position—to fully represent the original semantic information of the text. This system employs a stacked bidirectional Transformer model, with powerful text feature extraction capabilities that are more conducive to understanding semantics. BERT has shown powerful semantic extraction abilities, exhibiting the best performance among 11 different NLP tasks upon its release [60]. To understand the semantics of text, the context had to be obtained to fully grasp the

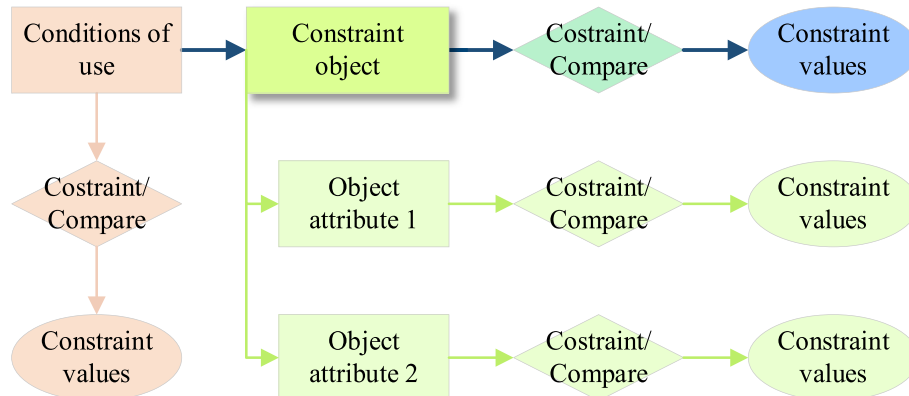


Fig. 1. General expression of specification.

meaning expressed in the text. Neural network models with memory ability have achieved good text semantic extraction results [28]. Long Short-Term Memory (LSTM) networks control the memory capabilities of neural networks through input, output, and forgetting gate units, and these networks are widely used in tasks with long-term data dependencies [61,62]. Extracting key information from the text would require a comprehensive understanding that incorporates contextual cues over long spans, a task well-suited to the capabilities of LSTM.

The second stage requires predicting the correct label for the extracted checking information, that is, determining which of the seven categories defined in Table 1 the checking information belongs to. Conditional Random Field (CRF) is a type of random field model that considers prior conditions, each output of the model considers the output status of adjacent positions, and uses the output of adjacent positions as prior conditions to make more reasonable predictions for the current position [63]. Such expression features exist in natural language expressions, such as: if the previous word is a noun subject, the probability of the current position being a copula verb or a predicate verb is higher, and the probability of being a noun is lower. The CRF network can predict the current output based on the output of adjacent positions by learning the state transition matrix between adjacent positions, thereby improving the prediction accuracy of checking information categories.

In this study, we first used the learning experience obtained by BERT training on large-scale datasets to encode the input text and extract text features. Then, the LSTM network was arranged bidirectionally to form Bi-LSTM to learn the text features of the preceding subsequent portions, utilizing the context memory capabilities of the Bi-LSTM network to deeply extract implicit entity knowledge in the text. Finally, the CRF model was used to learn the transition relationship between entity labels, combined with the neighboring label types, to predict the current entity label and improve prediction accuracy. This formed the checking information extraction network shown in Fig. 2, which sought to automatically annotate and extract key entity knowledge in the text.

3.2. Checking information parsing method

The checking information extraction network was used to remove useless information described in natural language, and scattered entity knowledge was obtained. The rule-constrained representation

recombined the entity knowledge to express complete semantics. Another Tool for Language Recognition (ANTLR), a syntax parser supporting grammar and semantic analysis, could flexibly customize syntax rules to handle language-based tasks [25,64]. As shown in Fig. 3, when parsing the text syntax, its lexical analyzer (Lexer) parsed the statement from character granularity into indivisible lexical units (Tokens), according to custom rules. Then, the lexical unit was passed to the parser (Parser) to build a syntax tree until the end of the string.

Based on the analysis results of the specification provisions and construction quality records, syntax parsing rules for checking information were developed, as shown in Fig. 4. Lexer rules must be represented in capital letters that correspond one-to-one to the entity labels defined in Table 1 to form a syntax tree based on the entity extraction results. Parse rules are represented in lowercase letters and can be nested with each other to simplify the expression. A syntax tree could have multiple branches, with the logical symbol “|” connecting different branches and the word behind “#” representing the abbreviation of the branch, “:” denoting the definition symbol, with the right side representing the definition value, “+” representing the front element appearing once or multiple times, “?” representing the front element appearing zero or one time, and “*” representing the front element appearing zero or multiple times, “-> skip” served as the built-in rule, indicating to skip the line content during parsing. For example, the “base” rule represented a basic constraint unit, with constraint word (req) + constraint value (val), which also considered the constraint value attributes that could appear before or after the constraint value.

The right side of Fig. 4 shows a parsing instance. After the original text was input into the checking information extraction network, the labeled entity was obtained. During the process of constructing the syntax tree, the entity with the label “obj” was first read as the “root” node, and then matched downward according to the Parser rule. From the Parser rule on the left, we observed that “root” was followed by one or more “types”, and “type” corresponded to 5 types of branching forms (i.e., #simple, #attribute). By traversing the entity tags, we determined that “obj” was immediately followed by “aobj + req + val”, and when two identical elements were next to each other, they merged and were treated as a single tag (e.g. the two “val” here). This did not conform to the first #simple rule of “type” (line 10 of the Parse rule) but conformed to the second #attribute rule (line 9 of the Parse rule); thus, it was recognized by the #attribute rule and treated as the first branch of the

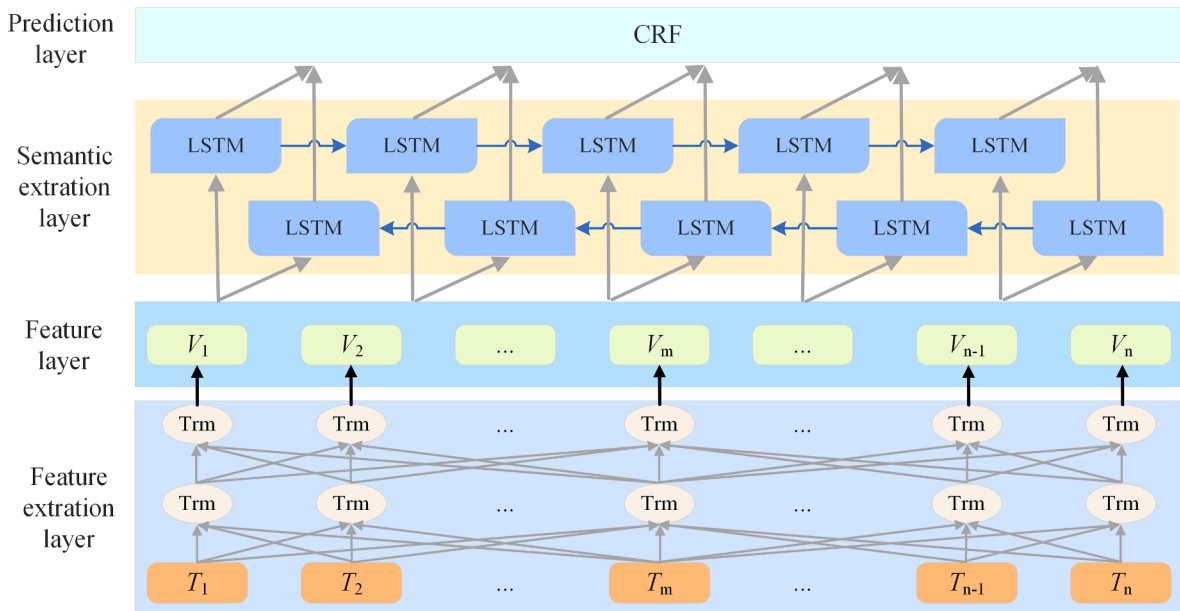


Fig. 2. Structure of the checking information extraction network, where Trm means Transformer, LSTM means Long Short-Term Memory, and CRF means Conditional Random Field.

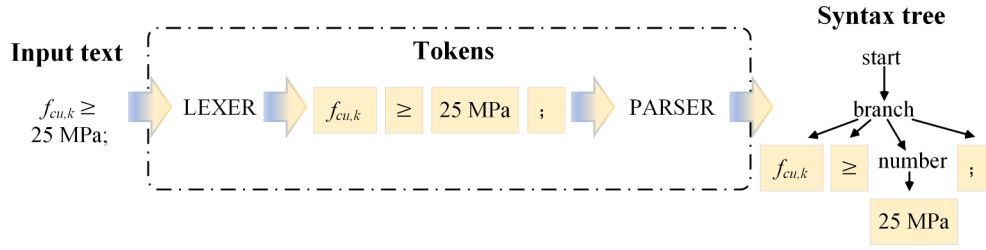


Fig. 3. Syntax tree construction process.

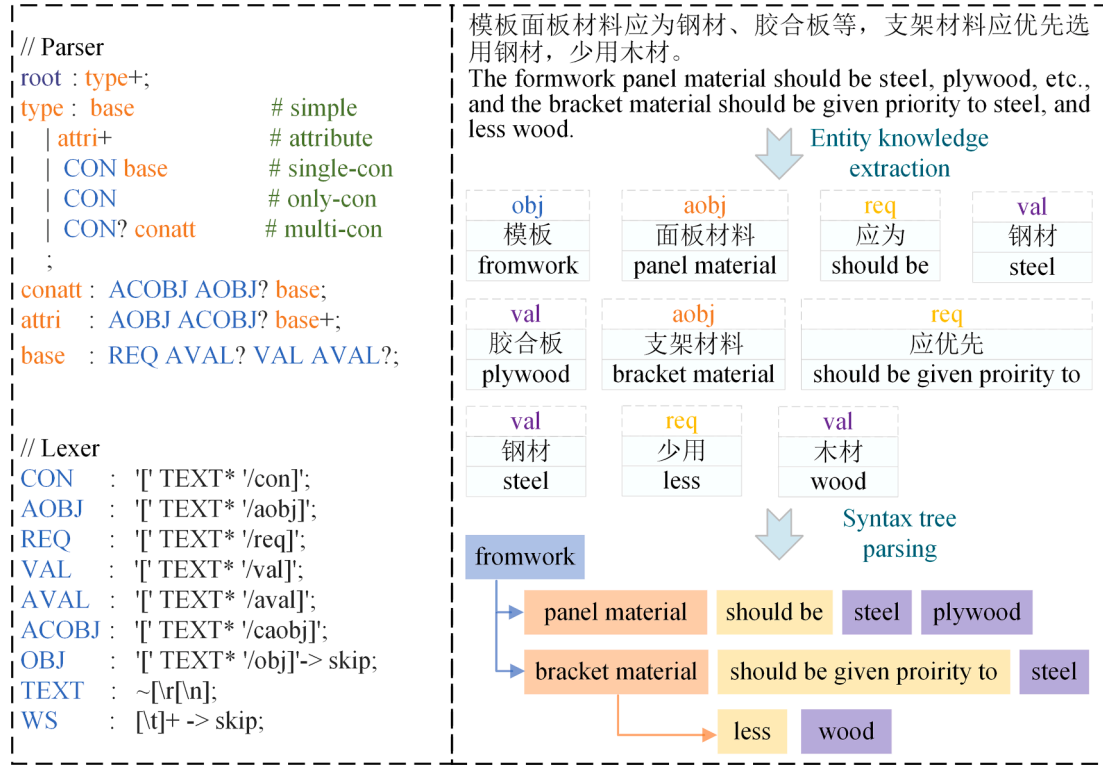


Fig. 4. Checking information parsing syntax rules.

syntax tree. The entity labels of each phrase were also prepared for the following scoring, and we continued to loop the above operations until all entity knowledge was formed into a syntax tree. After obtaining the syntax tree, mapping “req” to the corresponding mathematical representation was prepared for subsequent compliance checking, with the mapping rules example shown in Table 2.

3.3. Compliance check implementation process

3.3.1. Semantic alignment

After quality records and regulatory texts were parsed by the syntax

Table 2
Sample mapping rules.

Natural language descriptions	Mathematical description
Not greater than/less than or equal to/deviations are allowed...	≤
Greater than or equal to/should not be less than...	≥
Not equal to/avoid/strictly prohibited/should not appear...	≠
Is/was/should be adopted/met/followed/should conform...	=

analysis tool to obtain a syntax tree, the contents of the root nodes, branch nodes, and leaf nodes of the two syntax trees need to be aligned semantically to implement compliance comparison and scoring, forming a complete evaluation system. The root node was used to align the evaluation object, which determined whether the object was a required evaluation object specified by the specification. The branch node was then used to align the object attribute, determining the relationship between the recorded value of a certain attribute branch of the object and the regulatory constraint value. The most commonly used alignment methods, Boolean matching alignment and regular matching alignment [65], mainly involve splitting phrases into words and connecting them with logical connectors such as “and”, “or”, and “not”. When the words that makeup phrases fully match, these words can be considered aligned. This method serves as a simple and clear approach and the basis of many alignment methods. However, it also has obvious weaknesses, namely, it loses its judgment ability when the words cannot be completely matched.

Matching concrete dam construction specification nodes with on-site record nodes faces challenges due to potential discrepancies, making it difficult to align construction record phrases and specification clause phrases from only words. Semantic vector technology considers the textual context through the word embedding model and converts each word into a set of vector representations. Different phrases or character

combinations can be mapped to the same semantic space, using the distance function to measure the semantic distance of phrases in this space [66]. Similar phrases may exhibit similar contexts, and their semantic vectors will be closer. Unsupervised learning methods (e.g., Word2vec) can quickly obtain word vectors for each word based on the corpus context, but they also have obvious shortcomings: they require a large amount of corpus for training, but the domain dataset tends to be small. Moreover, for infrequent words (e.g., occurring fewer than 5 times), unsupervised learning methods are often used to delete the word directly, even though less common words may also convey crucial information [51]. Therefore, in this study, we used the SimBERT [67] word vector model with a deeper network hierarchy to obtain the semantic similarity of different words. The model adopted a twin BERT structure, feeding two phrases into one branch of the twin structure, which could obtain an accurate semantic vector representation of the phrases, and the semantic distance was calculated through the vectors. The model had the advantage of pre-training and running faster, which was favorable for the implementation of compliance detection. When aligning the specification nodes with quality record nodes, keyword matching was performed first, and if a match was found, direct comparison and evaluation could be carried out. With unregistered words, the word vector model could be used to calculate the semantic distance to determine whether the two were consistent descriptions.

3.3.2. Scoring design

Quality inspectors often have different recording habits, resulting in complexity and variations in quality records, with both quantitative and qualitative records in a single sentence. Hence, adopting more refined scores to indicate the degree of compliance could serve as a better approach. However, quality records and construction specifications frequently feature compound sentences, resulting in a syntactic parse tree that may contain multiple branches. As a result, a suitable process must be designed to identify the appropriate branch to score the quality

records. Therefore, this study designed a compliance scoring process, as shown in Fig. 5, based on the syntax tree and incorporated semantic alignment methods to realize the compliance scoring for construction quality.

The specific process corresponding to Fig. 5 could be described hereinafter. We utilized the syntax tree of the concrete construction quality record as the checking object and extracted a candidate specification syntax tree from the specification library. The target specification was determined by comparing the object nodes (obj) of the quality record with the objects in the specification text, and if the target specification was identified, the syntax tree branch comparison step was entered. The branch comparison process was as follows:

(1) If the branch length was 2, it was a “base” branch, and its “val” values of quality records and specifications were separately obtained for direct compliance checking.

(2) If the branch length was 3, if the first label was “con”, which semantically matched the “con” of the specification branch one by one, and the “val” of the two were compared when the specification branch matched. If the first label was “aoj”, the branch in the specification branch with the closest semantic distance to the “aoj” value was selected for comparison.

(3) If the branch length was greater than 3, then the branch contained multiple “base” combinations, and the branch was divided into segments based on the “val”. The operations in (1) or (2) were repeated for each segment.

To reflect the compliance degree of quality records in the form of scores, it was necessary to score the “val” nodes of each branch. The “val” nodes could be categorized into qualitative nodes and quantitative nodes, where the constraints of quantitative nodes were categorized into definite value constraints and range constraints. The semantic distance reflected the semantic difference between qualitative descriptions, and its range was only between [0,1]. Therefore, this study used the semantic similarity value as the qualitative checking score. The overall

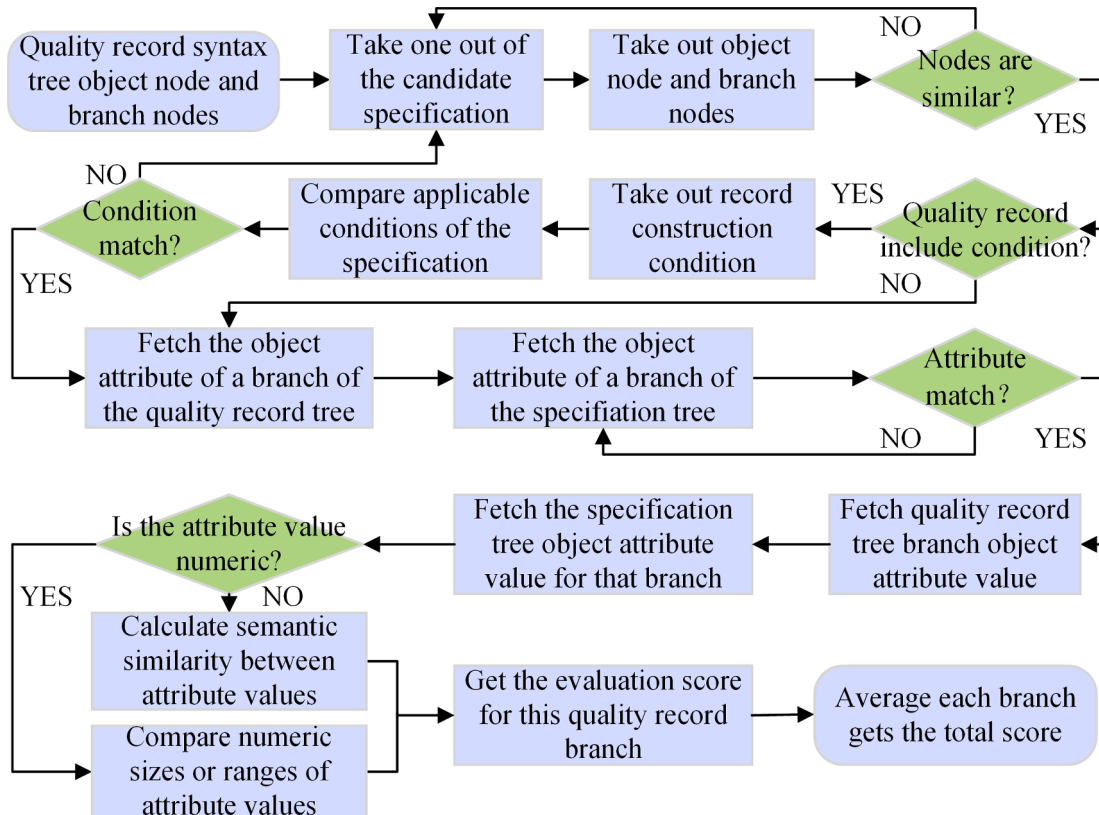


Fig. 5. Quality records compliance checking process.

formula for the “val” score could be defined by

$$S_{branch} = \begin{cases} 1, & \text{if } D \in D_{qualitative} \text{ and } V_r = V_c \\ 1, & \text{if } D \in D_{qualitative} \text{ and } V_r \in R_c \\ \text{dist}(\vec{m}_r, \vec{m}_c), & \text{if } D \in D_{quantitative} \\ 0, & \text{else} \end{cases} \quad (1)$$

where S_{branch} represents the branch score; D is the statement description; $D_{qualitative}$ and $D_{quantitative}$ represent quantitative and qualitative descriptions, respectively; V_r is the quantitative on-site quality record value; V_c and R_c represent the determined numerical class specification constraint value and range class specification constraint value, respectively; $\vec{m}_r$ and $\vec{m}_c$ denote the quality records qualitative description vectors and normative qualitative constraint vectors, respectively; and $\text{dist}(\bullet)$ is the calculation of semantic distance using cosine formula. The arithmetic mean of each branch score was taken to obtain the final score of the quality record, which could be defined by

$$S_{Total} = \frac{\sum_{i=1}^n S_{branch}}{n} \quad (2)$$

where S_{Total} is the final score of the quality record, and n is the number of branches contained in the parsing tree of the record.

3.4. Overall checking framework

By linking the four stages of (1) candidate specification retrieval, (2) checking information extraction, (3) checking information parsing, and (4) compliance checking implementation in series, this study proposed a basic framework for construction quality checking, as shown in Fig. 6.

(1) In the candidate specification retrieval phase, data input of the entire process was obtained, thereby starting the entire checking process. First, the prepared specification file was connected to the code framework, and each specification clause was separately rowed. Then we input the quality record text obtained from the construction site and

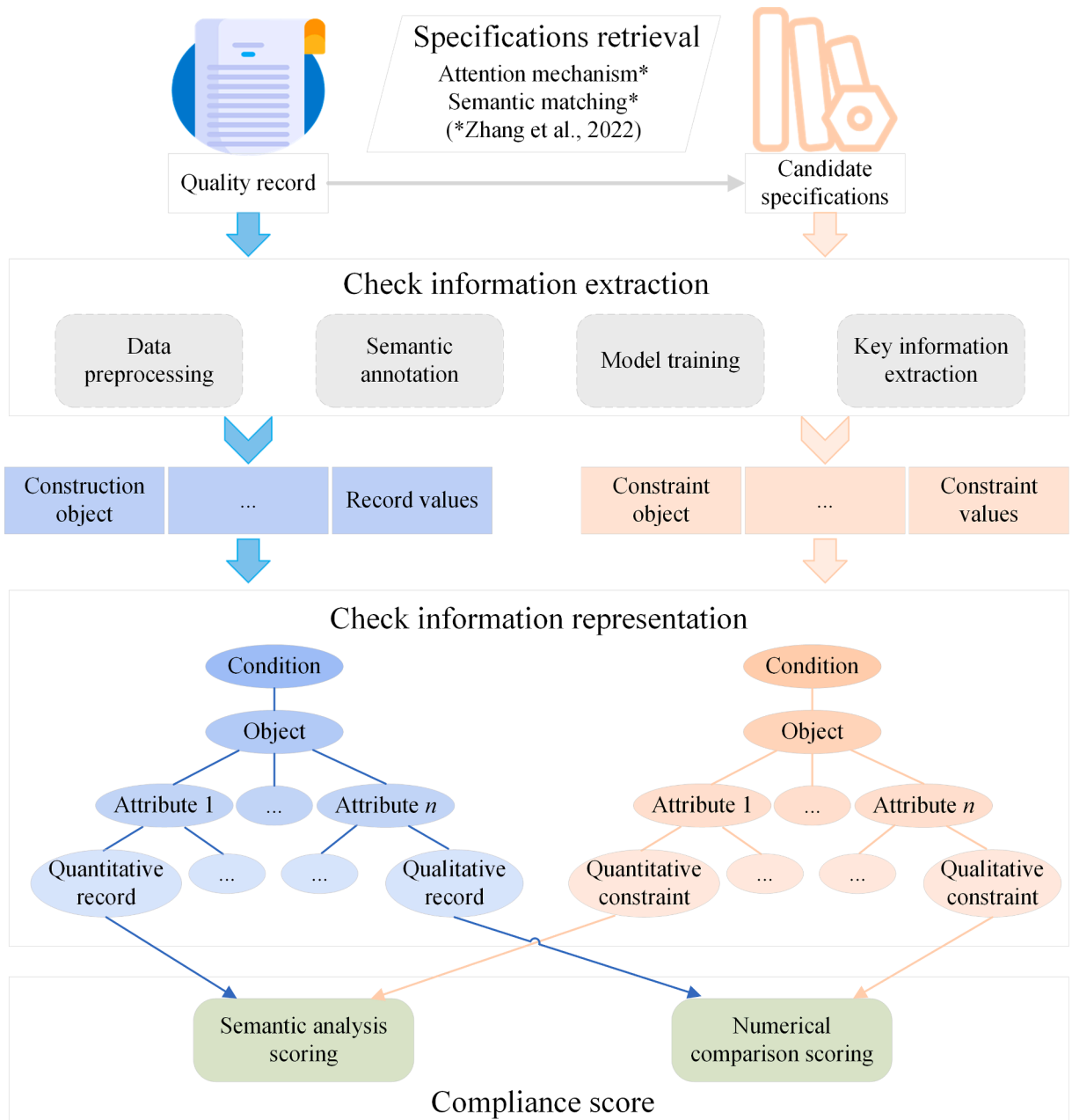


Fig. 6. Proposed framework for automated compliance checking of construction quality.

used the specification extraction method proposed by Zhang et al. [58] to extract the candidate specifications for each quality record from the specification file, forming a pair of sentences to be checked. This stage was not the focus of this study.

(2) In the checking information extraction phase, the sentence pairs obtained from the previous step served as the text input, and the deep learning algorithm was used to extract the check information. First, the text was preprocessed by word segmentation and stop word removal. Then, the text was input into the BERT model described in Section 3.1 to obtain the vector representation. Subsequently, the Bi-LSTM deep neural network was used to learn the sentence semantic information from the context vector. Finally, the CRF layer was used to automatically label the entity knowledge type of each phrase. According to the key entity types defined in Table 1, seven types of checking information, such as the constraint object, object condition, and attribute of object, were extracted to prepare for subsequent compliance checking.

(3) In the checking information parsing phase, the checking information obtained in the previous step was taken over and parsed the checking information into a syntax tree by customizing the parsing rules. According to the check information parsing syntax rules defined in Section 3.2, the ANTLR parser was used to form the extracted seven types of check information into a syntax parsing tree. The information parsing step could then parse a complex sentence containing multiple predicate verbs into a simpler sentence by branching, creating conditions for traversing each branch to perform compliance scoring.

(4) In the compliance checking implementation phase, the semantic alignment method proposed in Section 3.3.1 was used to check the compliance of quality records combined with the proposed scoring process. Semantic alignment was primarily used in the following aspects: to 1) determine whether the object node of the two parties (quality record and candidate specification) was a consistent description, and compare the branches of the syntax tree if they were; 2) determine whether the “con” node/“aobj” node of the two parties was a consistent description, and compare the “val” nodes if they were; and 3) calculate the similarity of the “val” node of the qualitative description class of the two parties as the compliance score of the qualitative constraint. According to the scoring process in Section 3.3.2, the final checking score was obtained, and the automatic compliance checking of

construction quality was realized.

4. Case study

In the following sections, the proposed framework’s effectiveness and performance were validated using an example of quality records generated during the construction of an actual concrete gravity dam. The Xiangjiaba Dam project, located in the Jinsha River section on the border of Yunnan and Sichuan Provinces in China, serves various purposes, including flood control and irrigation, while providing major hydroelectric power generation. Commencing in 2006, the dam was placed into operation for power generation in 2014, with a maximum dam height of 162 m, and a total concrete pouring volume of 11.14 million m³. The dam generated a large number of engineering construction records, and its project appearance and construction documents are shown in Fig. 7. The quality record texts generated during the concrete pouring phase of the project were taken as the test object, and two specifications referred to during the construction period, namely, “Specifications for hydraulic concrete construction” (SL 677-2014) and the concrete acceptance chapters of “Inspection and assessment standard for separated item project construction quality of water conservancy and hydroelectric engineering – Concrete works” (SL 632-2012), were used as the basis for conducting the case study. SL 677-2014 contained requirements and restrictions on materials, machinery, and operations for each construction process of hydraulic concrete, and SL 632-2012 contained constraints on various inspection indicators for concrete, serving as key materials for complete acceptance. Sample texts are shown in Table 3, where “mixed type” refers to texts containing both quantitative and qualitative descriptions.

4.1. Data preparation and gold standard setting

We established the gold standard for training the checking information extraction network. First, the corpus was divided into sentences as the smallest unit, and then the hydraulic engineering professional terminology library [68] was introduced to preprocess the sentences according to text segmentation, stop words removal, and uniform measurement units, to obtain a clean dataset. A total of 1,222 valid



Fig. 7. Xiangjiaba Dam project appearance and construction documents in Chinese.

Table 3
Text data samples.

Text type	Text statement
Quality records	Quantitative type 该部位采用无抗冻和侵蚀作用要求的混凝土, 其粗骨料吸水率为2%, 坚固性为3.5%. (original) Concrete without antifreeze and erosion requirements was used in this part, the water absorption rate of coarse aggregate was 2%, and the firmness was 3.5%. (translation)
	Qualitative type 监理测得在使用胶带机运送混凝土时布料均匀, 卸料处已设置挡板, 导管和刮板. (original) The supervision measured that the concrete was evenly distributed when the belt conveyor was used to transport the concrete, and the baffle, catheter, and scraper were set up at the unloading place. (translation)
	Mixed type 混凝土振捣时振捣器垂直插入下层5 cm, 留振时间设置合理, 无漏振, 在浇筑过程中实施全部检查. (original) When vibrating concrete, the vibrator was vertically inserted into the lower layer of 5 cm. The setting of the vibration retention time was reasonable and there was no leakage vibration. All inspections have been carried out during the pouring process. (translation)
Specification provisions	Quantitative type 7.4.6 基岩面和混凝土施工缝面浇筑第一层混凝土前, 施工方宜先铺一层2 ~ 3 cm厚的水泥砂浆, 或同等强度的小级配混凝土或富砂浆混凝土. (original) 7.4.6 Before pouring the first concrete on the bedrock surface or concrete construction joint surface, the construction side should first lay a layer of 2 ~ 3 cm thick cement mortar, or small-grade concrete or mortar-rich concrete of the same strength. (translation)
	Qualitative type 5.4.3 掺合料应储存到有明显标志的储罐或仓库中, 在运输和储存过程中应防水防潮, 并不应混入杂物. (original) 5.4.3 Admixtures should be stored in storage tanks or warehouses with obvious signs. They should be waterproof and moisture-proof during transportation and storage and should not be mixed with sundries. (translation)
	Mixed type 4.6.2.1 主控项目: 混凝土振捣时振捣器应垂直插入下层5 cm, 间距, 留振时间应合理, 无漏振, 无超振, 在混凝土浇筑过程应全部检查. (original) 4.6.2.1 Main control project: when vibrating concrete, the vibrator should be inserted vertically into the lower 5 cm, the spacing and the vibration retention time should be reasonable, there should be no leakage vibration and no super vibration, and all should be checked during the concrete pouring process. (translation)

clauses were obtained, of which 496 (40.59%) contained only one request word, and 726 (59.41%) contained multiple request words. In addition, the sentence lengths, the number of predicates, the number of leaf nodes (reflecting the average number of times each sentence had to be checked), and the proportion of grammatical nesting (the ratio of the number of sentences that reused parser rules to the total number of sentences) were counted, with the results shown in Fig. 8. For expert annotation, two authors and two engineers proficient in Chinese, with sufficient water conservancy professional knowledge background and a certain understanding of entity labeling, were selected. The entities were annotated according to the labels shown in Table 1. The first 10% was marked by all individuals, and the consistency of the labeling results was determined according to whether the F1 value of the labeling results reached 80% [69]. Subsequently, any inconsistencies were discussed by

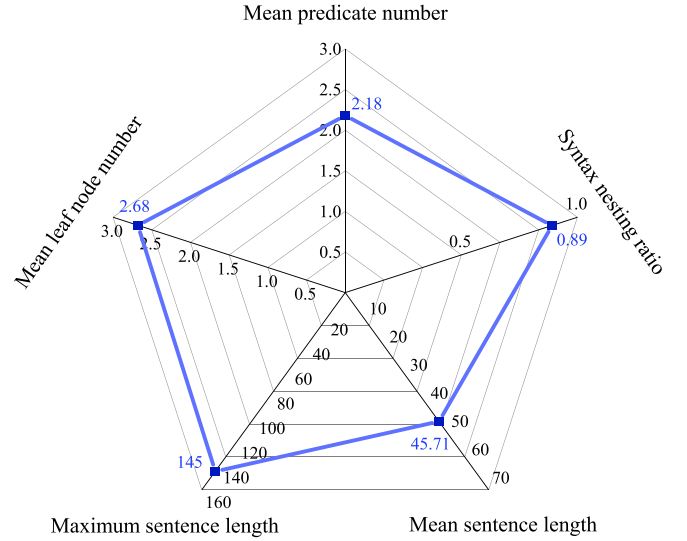


Fig. 8. Characterization statistics of the dataset used.

the experts to reach a consensus. The entire text was annotated, and the gold standard was obtained after verification by the authors. The annotation example is shown in Fig. 9, where “#o” represents non-entity knowledge. The corpus was divided into training sets, validation sets, and test sets according to the ratio of 6:1:1, and were used to train the checking information extraction network.

4.2. Entity knowledge extraction performance

Python 3.7 programming language was used to build the checking information extraction network framework for entity knowledge extraction. The network hyperparameters are shown in Table 4.

Precision, recall, and F1 were employed as evaluation metrics to assess the extraction effectiveness of the model, as described in Eq. (3):

$$\begin{cases} \text{precision} = \frac{\text{correct_recognize}}{\text{all_recognize}} \\ \text{recall} = \frac{\text{correct_recognize}}{\text{total}} \\ \text{F1} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \end{cases} \quad (3)$$

where *correct_recognize* indicates the number of positive examples that the model recognized correctly, *all_recognize* indicates the total number of positive examples identified by the model, and *total* indicates the number of all positive examples. Precision reflected the proportion of samples determined as positive by the model in the positive class, among all samples judged as positive by the model across all classes. Recall reflected the proportion of positive samples judged as positive by the model, among all positive samples, and F1 served as the harmonic mean of precision and recall, and could comprehensively reflect the model's performance. To verify the advantages of using the BERT model, this study gradually removed the pre-trained BERT module and CRF module and obtained (1) BiLSTM-CRF model and (2) BiLSTM-Softmax model as the control groups, with the Softmax function serving as one of the most commonly used classifiers in neural networks, and it could be used to build a baseline. The expression for Eq. (4) is given by:

$$f(x) = \frac{e^{x_i}}{\sum_{i=1}^n e^{x_i}} \quad (4)$$

where x_i is the i -th element vector, and n represents the feature dimension output by the last layer of neurons and refers to the number of final categories in the entity extraction task, transforming the feature vectors

Example 1:
模板#obj 面板材料#aobj 宜选用#req 钢材#val、#o 胶合板#val 等, #o 支架材料#aobj 应优先选用#req 钢材#val, #o 少用#req 木材#val。#o
The#o formwork#obj panel material#aobj should be#req steel#val, #o plywood#val, etc., and the#o bracket material#aobj should be given priority to#req steel#val, and#o less#req wood#val。#o
Example 2:
选用人工骨料时#con 施工方#obj 宜优先选用#req 石灰岩质#val 的#o 料源#aval。#o
When selecting artificial aggregates#con the construction party#obj should give priority to#req limestone#val sources#aval。#o
Example 3:
钢圆钢筋制成#con 的#o 箍筋#obj, #o 箍筋直径#con ≥#req 12 mm#val 时#o, 若#o 受力钢筋直径#aobj <#req 25 mm#val, 其#o 末端弯钩平直部分长度#aobj 应为#req 90 mm#val; 若#o 受力钢筋直径#aobj 为#req 28~40 mm#val, 其#o 末端弯钩平直部分长度#aobj 应为#req 105 mm#val。#o
Stirrups#obj made of round steel bars#con, When#o stirrups diameter#con ≥#req 12 mm#val, if#o the diameter of the stressed steel bar#aobj <#req 25 mm#val, #o the length of the straight part of the end hook#aobj should be#req 90 mm#val, if#o the diameter of the stressed steel bar#aobj is#req 28~40 mm#val, #o the length of the straight part of the end hook#aobj should be#req 105 mm#val。#o

Fig. 9. Example of entity knowledge annotation.

Table 4

Main hyperparameters of the neural network model.

Hyperparameter	Illustration	Value
LSTM_dim	LSTM hidden layer number	100
Pad_size	Length of each sentence after processing	150
Batch_size	The number of sentences entered each time	32
Dropout	Random inactivation ratio of neurons	0.5
Learning rate	Control gradient descent rate	0.001
Optimizer	Gradient descent optimization method	Adam
BERT_model	BERT pre-training parameters	BERT-Chinese

into predicted categories by dividing the elements into a probability distribution between 0 and 1, by taking the exponent of each element and dividing it by the sum of all element exponents.

The extraction results obtained by each model for the concrete dam construction specification entity knowledge verification set are shown in Fig. 10. By comparing the BERT_BiLSTM-CRF model with the BiLSTM-CRF model, we found that overall recall, precision and F1 value of the model using the pre-training strategy were 11.71, 13.93, and 12.89% higher. The improvement of the “aval” type was the most obvious, with values of 23.08, 25.19, and 24.14%. This was because the BERT model trained on a large-scale dataset could fully utilize the text representation patterns learned from the pre-training phase and transfer the learned experience to other fields. It could provide a more comprehensive representation of natural language text and more accurately extract text semantic features, thereby improving the performance of the BiLSTM-CRF model in information extraction. Comparing the BiLSTM-CRF model and the BiLSTM-Softmax model in the figure, the model precision using the CRF network was 5.32% higher, after comparing with the baseline model, and the average F1 value was 3.33% higher. This was because the CRF network could learn the transition relationship between adjacent labels and calculate the optimal label sequence using the associated features of adjacent labels, thereby improving the accuracy of checking information category prediction. Although the accuracy of the “aval” category was relatively low, it provided a supplementary description of the constraint value “val” and was considered minor information that had a limited impact on the overall compliance evaluation.

4.3. Concrete dam construction quality compliance checking performance

In this section, the initial focus was on presenting the computed

semantic distance results, which verified the superiority of the selected model by comparing the semantic distance and then illustrating the application effect of the proposed framework through engineering examples. In this study, the frequency of all words that appeared in the corpus was counted. According to the word frequency from high to low, six words were obtained by equidistant sampling with 500-word spacing, and a semantically similar word was obtained for each of the six words in the remaining words, with a total of 6 groups of 12 words obtained. Therefore, in the expected semantic distance results, the semantic similarity of the two words in a group should be higher and the similarity with other words should be lower. All words are renamed and ranked in alphabetical order, and the similarity was calculated using the Simbert network used in this study. The unsupervised Word2vec network was used as a baseline, with the results shown in Fig. 11, where the darker the color, the higher the similarity of words, and the closer the semantic distance.

As shown in Fig. 11, the semantic similarity calculated by Simbert achieved the expected results, with dark colors arranged near the diagonal, and the similarity within the group was higher, while similarity with other words outside the group was lower. However, the Word2vec model as the baseline had many calculation errors. For example, the similarity calculation value between A (mass concrete) and F (pollution) was 0.97, however, the two were almost unrelated. This proved that the Simbert network with a BERT model structure was more competitive in calculating the semantic distance of construction text.

To reflect the engineering application effect of the proposed framework, a stratified random sampling method was used to select 36 quality records from the dam construction site records, according to the proportion of each description type for checking. The positive cases consisted of records that met the specification requirements, and the negative cases consisted of manually set error records. Both record types included quantitative class checking, mixed class checking with both quantitative and qualitative descriptions, and qualitative class checking. The overall scoring results are shown in Fig. 12, with green indicating records with a score greater than 0.5 and orange indicating records with a score less than 0.5.

As shown in Fig. 12, a total of 36 records were tested, except for three records that were not scored, and the 18th record was not correctly scored, resulting in a test accuracy of 88.89%. Checking required 371 s, with an average time of 10.3 s per record, indicating that the proposed method had satisfactory usability. In most positive cases, the scores were greater than 0.5. In the manually set negative cases, most of the model scores were less than 0.5, and the 30th record contains two branches,

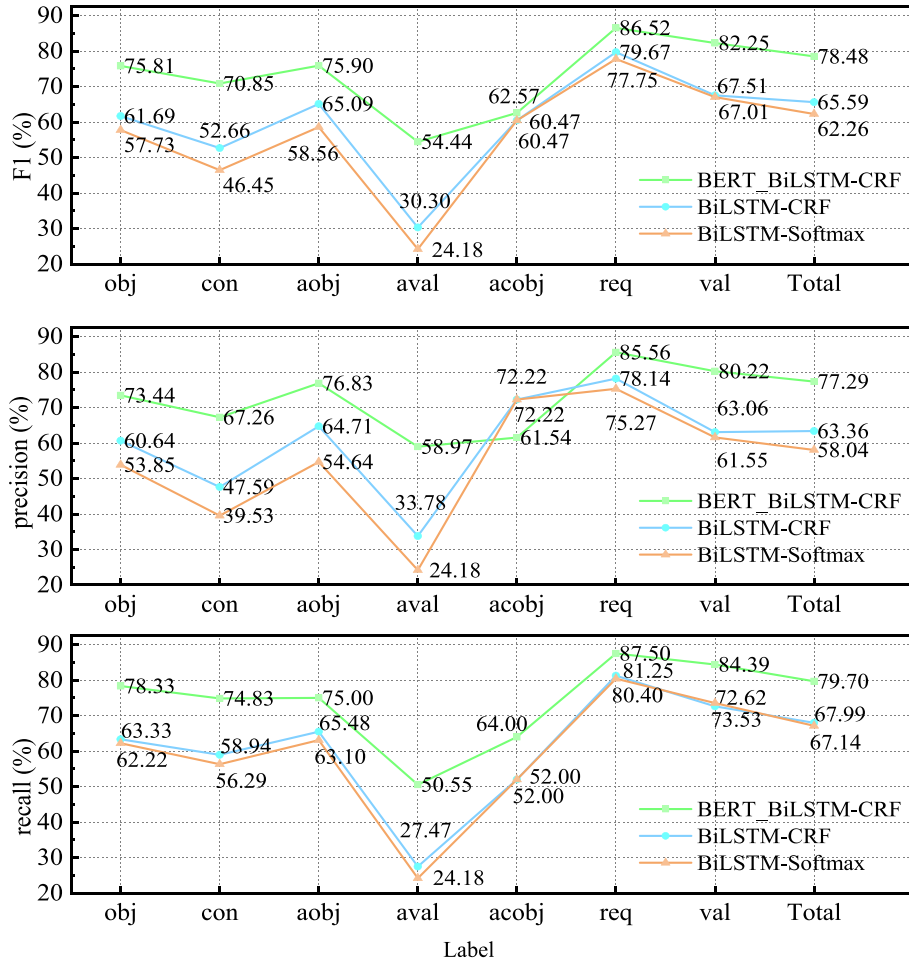


Fig. 10. Entity knowledge extraction results.

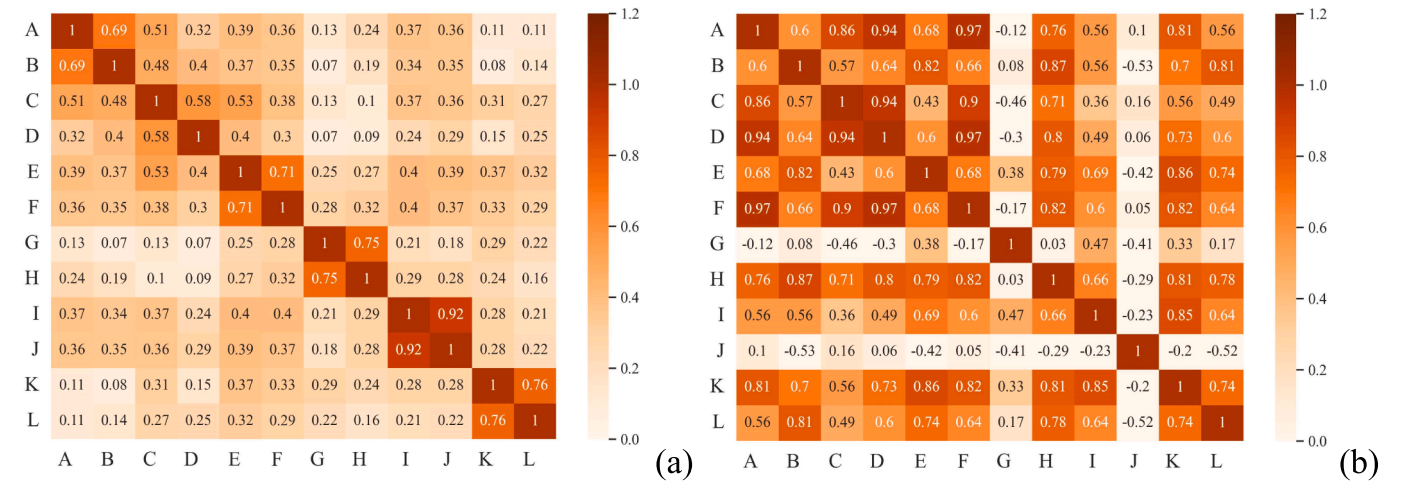


Fig. 11. Semantic similarity heatmap: (a) Simbert network results, and (b) Word2vec network results, where A is mass concrete, B is roller compacted concrete, C is expansive cement, D is ordinary Portland cement, E is oil stain, F is pollution, G is the number of detections, H is detection frequency, I is rainproof measures, J is rainproof facilities, K is unobstructed, and L is unblocked.

with an incorrect value set for only one branch, resulting in a score of 0.5. A few typical results are shown in Table 5 in detail, and a portion of the analysis process and the checking results are shown in Fig. 13.

Table 5 shows that for relatively regular specifications and records, the proposed method could accurately implement compliance checking according to the specification constraints. The method presented

accurate checking effects for single-digit constraints (tests 1 and 7) and constraints with a certain range (tests 13, 29, and 30). For mixed quantitative and qualitative constraints (test 13 “layer by layer from bottom to top” and “0.2 ~ 0.5 MPa”), the method could perform checks from both semantic and numerical perspectives. For qualitative constraints (tests 21, 23, and 34), the method could provide correct

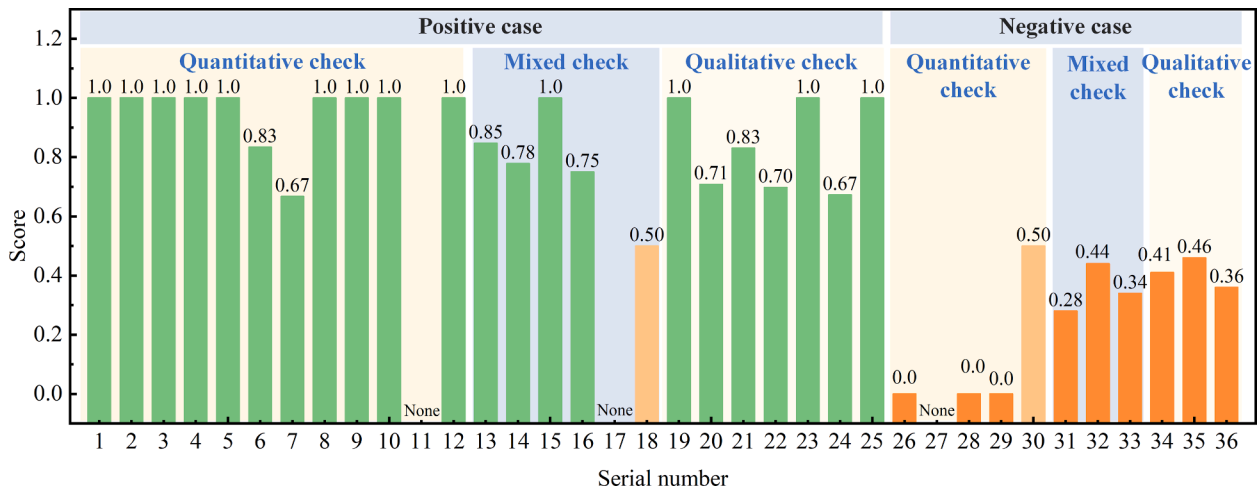


Fig. 12. Compliance check scoring results.

checking results from a semantic analysis perspective. For manually set incorrect values (tests 29, 30, and 34), the method provided lower scores to attract more attention from quality management personnel. As shown in Fig. 13, quality records were input and the specification text was constructed with similar syntax trees through syntactic parsing. By aligning the semantics of corresponding nodes, the specification constraint branches were found as evaluation criteria. By implementing the process shown in Fig. 5 at the branch level, the on-site records could be checked and scored, which could assist concrete dam quality managers in quickly screening potential non-compliant operations. This could prove helpful for managers to quickly discover problems and make corresponding decisions, thus, ensuring the quality of dam construction and improving the corporate profits of construction units.

4.4. Error analysis of checking results

Error analysis was performed on cases of test errors to determine the source of the errors. Insufficient syntax rules and missing information in the syntax tree were some of the reasons for the errors. Three samples (11, 17, and 27) could not be scored, as shown in Fig. 12. After analysis, we found that the specification for test 11 was very complex, with 9 branches, including 4 conditional branches, which exceeded the parsing scope of the syntax tree rules and caused the compliance check to fail. In test 17, the description “air content should be tested once every four hours” and the dense presence of two “val” constraint items caused the syntax parsing to fail. Therefore, a method to improve these errors would involve developing targeted parsing rules and compliance scoring rules.

The accuracy of entity knowledge extraction served as another cause of the error, and phrases located in different positions had to be identified as different entities. For example, “air content” at the beginning of a sentence was likely to be “obj”, while in the middle of a sentence, it was likely to be “aobj”. This resulted in recognition pressure to the entity knowledge extraction model. Due to the low accuracy of “aval”, it could lead to errors in entity extraction during use, with the wrong order of entities potentially leading to the selection of the wrong parser rules, and forming the wrong syntax tree or preventing the syntax tree from forming. The wrong entity could also cause errors in semantically aligned nodes, resulting in errors in the model score. For example, in test 18, two branches should have been checked, but the model only scored one of the branches, due to the incorrect identification of a branch node. After the average operation of the branch scores, the final score was 0.5 points. The method to refine these errors would involve augmenting the training dataset or developing a more potent deep-learning model.

Certain errors were also present in implementing semantic alignment

and qualitative checks using semantic distance based on word embeddings. Word embeddings were obtained through neural network language models trained on large-scale datasets, which served as a better way to represent text semantics. However, this also included some shortcomings. Such as the vector effect was limited by the training data, and the recognition of antonyms often found in similar contexts was not accurate. For example, there were some cases where the score was not 1 point in the compliance assessment containing qualitative rules.

5. Discussion

Concrete dam projects are large in scale, involving many processes over a long construction period. Quality records on the construction site may be highly scattered and disordered, hindering the speed and accuracy of managers to determine quality compliance. This can lead to a contradiction between the low efficiency of project quality management and the pursuit of high profits by the enterprise. For a large number of concrete dam construction quality records, relying on the experience of quality management personnel to determine whether the construction quality meets the specifications carries significant risk. The process of reviewing specification clauses based on quality records is time-consuming and labor-intensive and may be prone to fatigue and mistakes, resulting in engineering defects and, in severe cases, posing a significant threat to life and property. The automatic compliance check method proposed in this study for concrete dam construction quality offers a fast checking and scoring solution. Managers would only need to input on-site quality text records, and the method would screen the most valuable normative clauses in a given normative library. In this method, a checking information extraction network can be used to extract key entity knowledge, using a syntax analysis algorithm to build a knowledge syntax tree, and finally comparing the syntax tree constraint values to obtain a dam construction quality compliance score. The accuracy of the check reached 88.89%, and the average checking time per record was 10.3 s, which could significantly reduce the checking range of management personnel and effectively improve the efficiency of engineering quality management. The primary advantages of this study were as follows:

On the theoretical level, this study proposed a new method for the automatic compliance checking of on-site construction quality for concrete dams, enhancing the theoretical system of automatic compliance checking. (1) This method extracted dense normative knowledge information through the proposed deep neural network, which had excellent semantic feature extraction and text context memory capabilities, and it still demonstrated good performance in the face of complex concrete construction specification texts with different semantics in

Table 5
Examples of compliance checking.

Serial number	Quality records	Normative basis	Score
1	Concrete without antifreeze and erosion requirements is used in this part, the <i>water absorption rate of coarse aggregate is 2%</i> , and the <i>firmness is 3.5%</i> .	5.3.6.5 The quality of coarse aggregate: the <i>water absorption rate</i> of concrete without frost resistance and erosion <i>should be $\leq 2.5\%$</i> ; the <i>firmness should be $\leq 12\%$</i> .	1.0
7	After testing, the capacity of the on-site mixer is 0.8 m^3 , the <i>maximum aggregate particle size is 80 mm</i> , and the <i>mixing time of the self-falling mixer is 120 s</i> .	7.2.7.1 $0.75 \text{ m}^3 \leq$ mixer capacity $\leq 1 \text{ m}^3$, the <i>maximum aggregate particle size is 80 mm</i> , and the <i>minimum mixing time is 90 s</i> when a self-falling mixer is used.	0.67
13	The process of grouting procedure is to rise layer by layer from bottom to top without interruption, the <i>grouting pressure is 0.3 MPa</i> , and the <i>rising speed of the slurry is 0.6 m/h</i> .	7.6.5.3 The grouting procedure should start from bottom to top and gradually rise without interruption. The <i>grouting pressure should be 0.2 ~ 0.5 MPa</i> , and the <i>rising speed of the slurry should be 0.5 ~ 1 m/h</i> .	0.85
17	The air content of concrete mixed with an air-entraining agent is set to be tested every 4 h , and the <i>average deviation is 0.5%</i> .	11.3.4 The air content of concrete with an air-entraining agent should be tested once every four hours , and the <i>allowable deviation of air content is 1.0%</i> .	None
18	The supervisor measured the <i>surface moisture content of the fine aggregate at the mixing site to be 3%</i> , and accelerated dehydration measures were adopted when necessary.	5.3.5.2 The <i>surface moisture content of fine aggregate should not exceed 6%</i> , and should be kept stable. If necessary, measures to accelerate dehydration would be taken .	0.5
21	The supervision carried out the general item checking and found that the metal water stop was welded on two sides ; the test method is observation ; the number of inspections is any welding point .	4.5.2 General items: metal water stop should be double-sided welding ; the test method is observation ; the number of inspections is per weld .	0.83
23	The supervisor measured that the underwater concrete part was made of pozzolanic Portland cement .	5.2.1.3 For internal concrete, underwater concrete and foundation concrete, medium-heat Portland cement, low-heat Portland cement and ordinary Portland cement should be used, and pozzolanic Portland cement should also be used.	1.0
29	The supervisor measured the speed of the <i>mixing drum to maintain a speed of 10 r/min</i> during the delivery of the mixer truck.	7.3.7.3 During the transportation of the mixer truck, the <i>mixing drum should keep a slow rotation of 3 ~ 6 r/min</i> , and should not add water to the mixing drum.	0.0
30	The cooling water pipe is used to control temperature on site, the difference between the concrete temperature and the <i>water temperature is measured to be 25°C</i> . After measurement, the <i>water flow rate in the pipe is 1.2 m/s</i> .	8.3.3 When using cooling water pipe temperature control, the difference between the concrete temperature and <i>water temperature should not exceed 25°C</i> , and the <i>flow rate of the water in the pipe should be 0.6 ~ 0.7 m/s</i> .	0.5
34	Brush quality checking of the release agent, the quality of brushing is significantly different in color depth ; the checking method is to view	4.3.2.17 Release agent brushing quality should meet the standards, uniform brushing, no obvious color difference ; the checking	0.41

Table 5 (continued)

Serial number	Quality records	Normative basis	Score
	the brushed template; the number of tests is sampling checking.	method should consist of consulting the product quality inspection certificate and observations; the number of tests should be all.	

Note: Bold denotes qualitative checking, while italic denotes quantitative checking.

different positions. Therefore, the specification knowledge could be expressed more abundantly. (2) The method used syntax tree parsing algorithms to organize specification knowledge information into tree structures without complicated rulemaking. This method was open-source and supported user-defined parsing syntax rules, making it highly extendable for different task objectives. (3) The proposed method utilized the semantic distance alignment of word embedding vectors to align quality records with the syntax tree branch nodes detailed by specification regulations and could evaluate the qualitative compliance degree of specification regulations. This expanded compliance checking scope for quantitative constraints compared to previous methods, serving as a beneficial exploration for the implementation of automatic compliance checking for the entire life cycle of dam design, construction, and operation.

On the practical level, this proposed method had a stronger integration ability and automation level, which could assist concrete dam construction quality managers in discovering construction operations that violate the specifications in time and improving the efficiency of construction quality management. (1) The proposed method could make the quality compliance checks more flexible. This method was developed in the same environment for ease of integration, supporting custom candidate specification libraries and making it convenient to customize and update specification information with strong flexibility and integration capabilities. For different checking processes or even different construction projects, the quality compliance checking of new projects could be assisted by changing the national policy documents in the specification library and by fine-tuning, making it flexible and convenient. (2) The proposed method made the workflow of quality managers more efficient. Without the assistance of the proposed method, quality inspectors would need to read national normative documents item by item, read the content of normative constraints and on-site quality inspection results one by one, and then make compliance scores, resulting in a time-consuming, labor-intensive, and inefficient process. With the assistance of the proposed method, managers would only need to input the quality text records obtained from on-site testing, and the system would automatically complete a series of operations such as specification retrieval, text preprocessing, entity knowledge extraction, knowledge syntax tree construction, and normative constraint comparison, resulting in a high level of automation throughout the checking process. This would save time and money, free up labor, and allow managers to focus on more critical engineering decisions. (3) The proposed method made quality compliance checks more accurate and transparent. This method could not only quickly obtain a compliance score but also provide the original text of the specification as scoring for the reference of managers. The scoring of each construction record had a basis and traces, which enhanced the credibility of quality compliance checking and provided a normative reference for possible quality accountability in the future. Overall, this method could quickly narrow the scope of quality compliance checking, assist managers in controlling the construction quality of concrete dams, save on manpower and material resources for dam construction quality management, and provide useful explorations for intelligent construction quality management.

Although this study was based on Chinese text, this proposed framework could be applied to multiple language texts, as the NLP

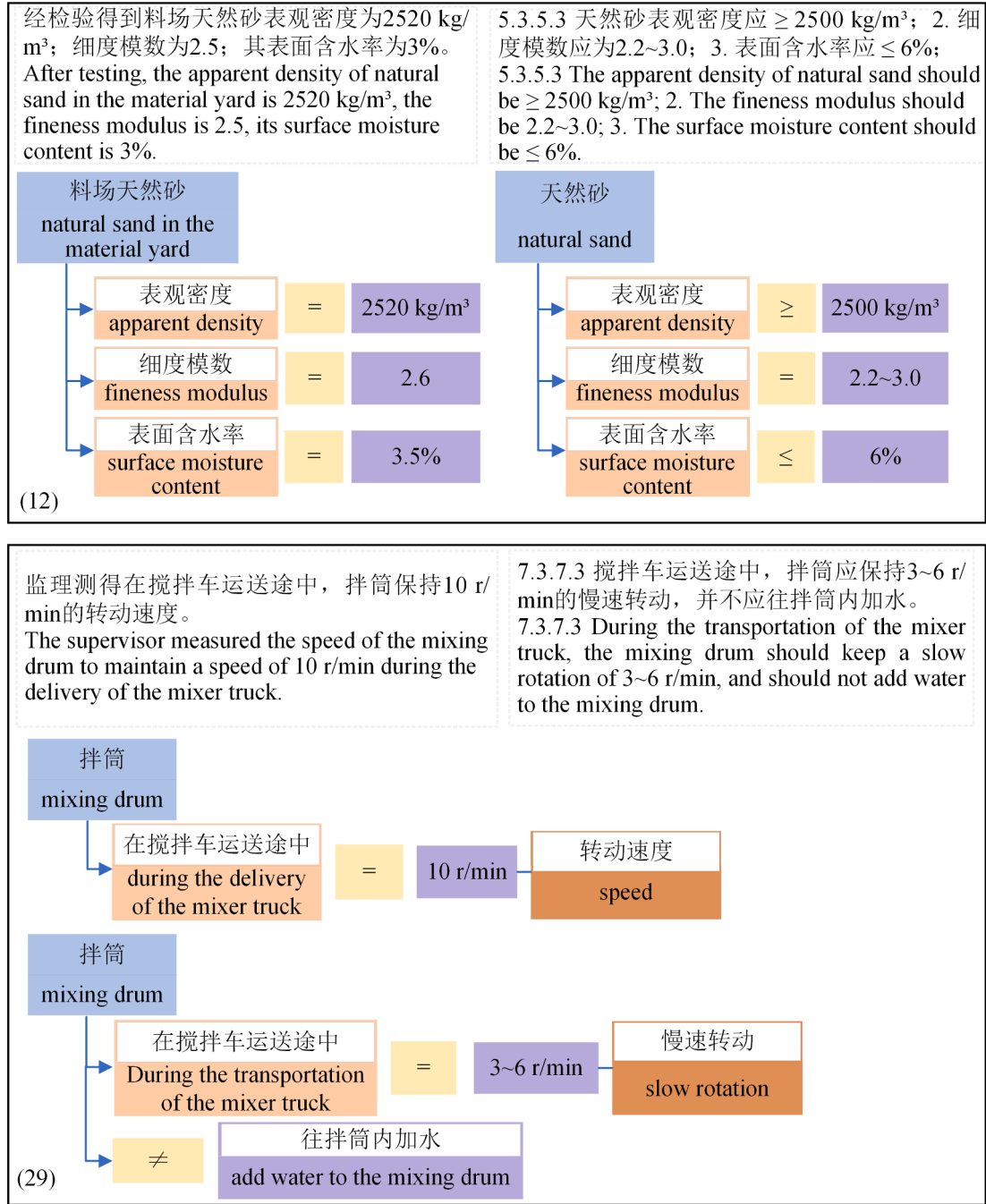


Fig. 13. Syntax analysis results: (12) score = 1.0, and (29) score = 0.0.

technology used in this study has made significant progress in a variety of text backgrounds. For example, entity recognition technology has been used on English datasets [28,58]. The syntax parsing and scoring design methods were language-insensitive, and the parser rule would only need to be fine-tuned according to English expression habits. The parser rule only contained a few lines (Fig. 4), with relatively simple modifications. Only the semantic alignment model would need to be adjusted and trained using English datasets (existing models, such as Sentence-bert [70], or self-training new models). Overall, this proposed framework had a small adjustment cost, allowing it to be applied to multiple language texts.

6. Conclusions and future work

Accurately and efficiently realizing the automatic compliance checking of the concrete construction process is critical in ensuring the quality of concrete dams. This study proposed an automated quality compliance check framework for complex on-site concrete construction text records and normative expressions. We then verified the applicability of the proposed method using an example of actual construction quality records. The main contributions were as follows:

(1) A concrete dam construction quality compliance automated check framework was proposed, which combined syntax parsing with semantic distance. Because construction quality checking mostly depends on plain text records, this framework used deep learning technology to extract the key entity knowledge of the text, using the syntax

analysis rules to reorganize the entity knowledge into a knowledge syntax tree according to branches. With semantic similarity as the criterion, this strategy aligned quality records nodes with specification syntax tree nodes and implemented compliance checking by comparing syntax tree branches. We then used a finer-grained scoring method instead of binary compliance/non-compliance to represent the degree of compliance, which was more suitable for construction text records with multiple branches. The accuracy of the test was 88.89%, demonstrating the engineering applicability of the proposed method.

(2) A complex text parsing model based on entity knowledge extraction and syntax parsing was constructed. The BERT pre-training network was introduced to improve the BiLSTM-CRF model, which enhanced the text semantic extraction ability of the model, the experiment showed that its F1 value was 12.89% higher than similar models. The syntax parsing algorithm could rebuild the extracted entity knowledge into a logical relationship knowledge syntax tree, as well as parse complex text into simple branches, offering flexible scalability and a powerful text representation ability.

(3) A qualitative compliance checking method based on word embedding vector semantic distance was proposed. For qualitative norm constraints that would be difficult to represent numerically, this study combined NLP technology to use the word vector to represent qualitative constraints. The vector semantic distance was used to reflect the compliance degree of qualitative description using the feature where semantic similarity description had a closer vector semantic distance, which expanded the theoretical system and application scope of compliance checking.

This study also had some limitations, and efforts will focus on the following aspects in the future. (1) We will collect more normative texts, expand the representation method of general features of norms, collect additional construction site records data, and broaden the scope of application of the proposed method. (2) Due to the randomness of text description, the existing parsing rules could not cover all situations, leaving room for improvement in handling complex qualitative descriptions of logic. This will need to be developed to improve syntax parsing rules and enhance the text parsing and representation capabilities of the method, as well as improve the coverage of syntax parsing. (3) Higher precision entity knowledge extraction models will need to be trained. In the face of complex text descriptions, entity knowledge extraction accuracy will have a large impact on the results, requiring better models to accurately label knowledge categories and improve the overall framework accuracy.

CRedit authorship contribution statement

Qiubing Ren: Writing – review & editing, Project administration, Conceptualization, Supervision. **Dongliang Zhang:** Methodology, Software, Writing – original draft, Writing – review & editing. **Mingchao Li:** Formal analysis, Funding acquisition, Resources. **Shu Chen:** Supervision, Resources. **Dan Tian:** Conceptualization, Writing – review & editing, Methodology. **Heng Li:** Conceptualization, Funding acquisition. **Leping Liu:** Data curation, Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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